

Simulating Logistics Systems

Lesson 9/15

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Outline of this lecture

Terminal and Non-Terminal Systems

Verification and validation

Sizing simulation runs

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Terminal and Non-Terminal Systems [Banks, 2005, Law et al., 2007, Pidd, 1998]

- ▶ **Terminal systems:** Restarts periodically. The system's operating time must match the simulation run duration (e.g., 6 hours, 10 a.m. to 4 p.m.) and start with no resources. The transient period is part of the system.
- ▶ **Non-terminal systems:** system does not restart.

Examples:

1. **Terminals:** Bank agencies, restaurants, and services that don't operate 24/7.
2. **Non-terminals:** port, mining, steel mill, hospital, etc, operating 24/7.



Figure: Queues in bank

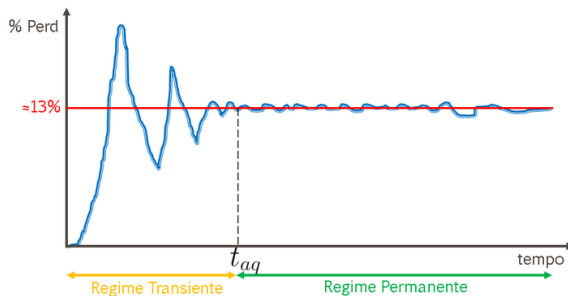


Figure: Queues in hospital

Transient and Steady-State Periods

What are the transient and steady-state (permanent) periods of a stochastic regime?

Example of a Telephone Center ($t_{aq} \equiv$ Tempo de Aquecimento (Warm up))



Obtaining good estimates (Note: *Make numerous samples from observations*):

- ▶ Simulate a very long period (but this consumes computing resources)
- ▶ **Eliminate** the transient period
- ▶ Start the system already in a steady state (unworkable)

Simulation Runs

- ▶ **Simulation Run** = n replications. Run results for a variable x must be expressed with **statistical confidence** and **precision**.
- ▶ **Confidence**: Confidence interval (CI): $P = 1 - \alpha$ (e.g.: 95%)
- ▶ **Precision**: Width or size of the interval: $(\bar{x} - h, \bar{x} + h)$

$$P(\bar{x} - h, \bar{x} + h) = 1 - \alpha$$

μ $\equiv$ Expected value

$\bar{x}$ $\equiv$ Sample mean

s $\equiv$ Sample standard deviation

$h = t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \equiv$ Half-width of CI, i.e. *precision*

$t_{n-1, \frac{\alpha}{2}} \equiv$ t-Student distribution with $n - 1$ degrees of freedom and α significance

Simulation Runs

For e.g., the Telephone Center, 1 simulation run with n replications

- ▶ For each replication i : $\bar{p}_i = \frac{\text{Lost calls}}{\text{Calls}}$
- ▶ Final result (CI): $CI = \bar{p} \pm h$

Where:

$\bar{p} \equiv$ Mean of lost calls

$h = t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \equiv$ Half-width of CI, i.e. *precision*

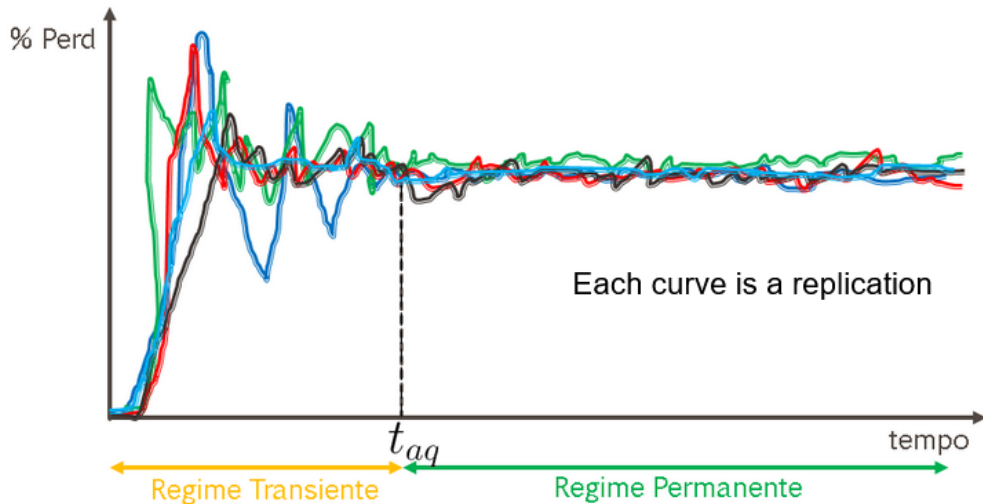
$t_{n-1, \frac{\alpha}{2}} \equiv$ t-Student distribution with $n - 1$ degrees of freedom and α significance

We can **reset** the system and/or statistics between replications.

- ▶ **System reset** restores variables to their initial values and resets the system to its initial state. *Each replication involves a new warm-up period.*
- ▶ **Statistics reset** to zero with each replication. *Data collection for statistics is restarted with each replication.*

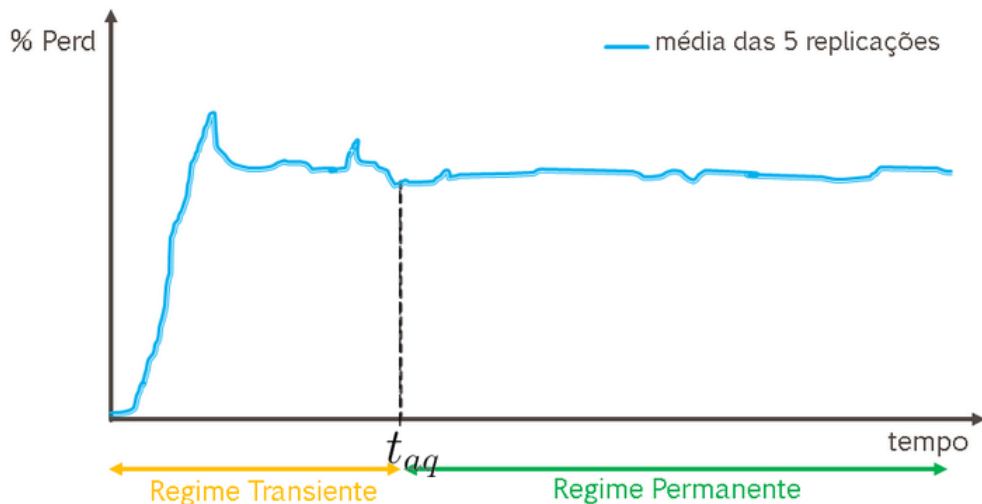
Warm-up period

Set the warm-up period by repeatedly observing the system's stability.

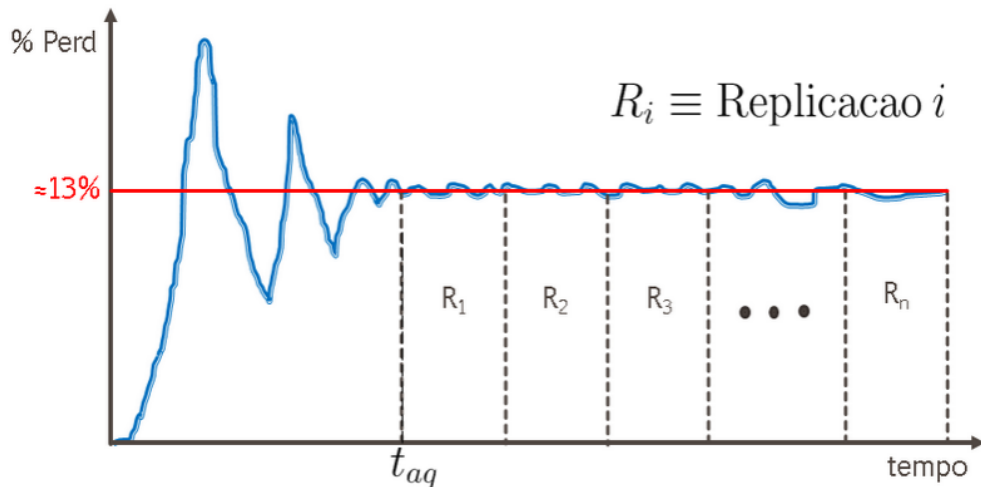


Warm-up period

Averaging the replications makes observation easier. Calculate a moving average with m values from the average of the five replications.



No-restarted statistics



Outline of this lecture

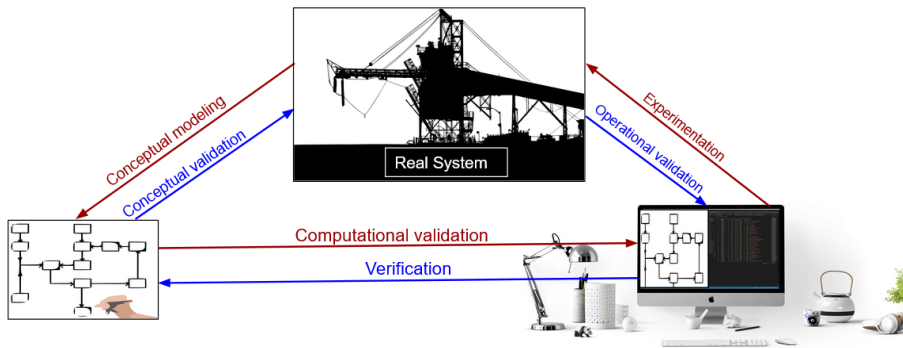
Terminal and Non-Terminal Systems

Verification and validation

Sizing simulation runs

Verification & validation [Banks, 2005, Law et al., 2007, Pidd, 1998]

Verification and **validation** are key to a successful simulation study. They should be used throughout the entire project.



- **Validation (conceptual model):**
Are we developing the **right model**?
(*detail level, scope, represent the system?*)

- **Verification (computational model):** Are we developing the model **correctly**? (*bugs, right computations.*)

Verification & validation

Verification techniques

- ▶ Modular implementation and check
- ▶ Simple input: Manual calculations
- ▶ Group review
- ▶ Animation: visual perception of errors
- ▶ Debugging tools for step control
- ▶ Manual simulation for simpler models

Before validation...check Error Types

- ▶ I: Model is **valid** but *rejected*.
- ▶ II: Model **not valid** but *accepted*.
- ▶ III: Model *deviates* from **objectives**.
- ▶ **Data!**: Check data quality and outliers (*garbage in, garbage out*)

Validation techniques

- ▶ Turing test (black-box validation)
- ▶ Sensitivity analysis
- ▶ Face-to-face validation
- ▶ Model replication
- ▶ Compare with previous right models



Level 1 (Qualitative)

- ▶ Does the model behave like the real system?
- ▶ Are there long queues in the model where there are long queues in reality?
- ▶ Where there are no queues in the model, are there no queues in reality?
- ▶ Is the bottleneck machine in the model the same as the actual bottleneck?
- ▶ Is the most-used resource in the model the same as in reality?
- ▶ Is the least-used resource in the model the same as in reality?

Answer "yes": The model has **qualitative validity**. Proceed to next validation level.

Answer "no": **Reject** (*invalid model*).

Level 2 (Informal Quantitative): " Compare a few samples of data with the actual system to validate the *order of magnitude* of the parameter." Example:

- ▶ A queue should have a *maximum of 5 people*. If the average in the model is 0 or shows 20 people, then the model is **invalid**.
- ▶ The daily production is *approximately 150 pieces*. If the model produces 400 pieces, then the model is **invalid**.
- ▶ Operator utilization is *approximately 70%* in a work shift. If the model generates utilization *ranging from 60% to 80%*, it can **proceed to the next level** of validation.

Level 3 (Formal quantitative): *"Established statistical techniques for validating various performance indicators."*

- ▶ Inference test on two independent sample means: *For a given performance measure, n replications of a simulation are compared with n replications of actual results. The test shows whether the simulation results are equivalent to the actual results for a given level of statistical confidence.*
- ▶ **Example:** "Online orders in e-commerce." *Performance measure:* The percentage of orders fulfilled (service level) is defined as the number of orders delivered on time divided by the total number of orders.

Operational Validation

Example: Online orders in e-commerce

► Real vs Simulation results

Replicacao	Dados Reais	Modelo de Simulacao	Diferenca (real - simulado)
1	93%	94%	-1%
2	94%	97%	-3%
3	96%	97%	-1%
4	98%	96%	2%
5	96%	97%	-1%
6	99%	95%	4%
7	98%	96%	2%
8	94%	96%	-2%
9	95%	94%	1%
10	99%	96%	3%
Media	96.2%	95.8%	0.4%
Desvio-Padiao	2.2%	1.1%	2.3%

We calculate the confidence interval (CI) for the difference between the two systems (real vs. simulated) using the same number of data points. If the CI includes 0, the responses are equivalent with a confidence level of $100(1 - \alpha)\%$.

$$\bar{x}_s - \bar{x}_r \pm t_{2n-2, \frac{\alpha}{2}} \sqrt{\frac{S_s^2 + S_r^2}{n}}$$

$\bar{x}_s$ $\equiv$ mean of simulation result

$\bar{x}_r$ $\equiv$ mean of real result

$\bar{S}_s$ $\equiv$ std. dev of simulation

$\bar{S}_r$ $\equiv$ std. dev of real

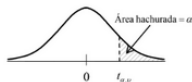
n $\equiv$ data (same for real and sim)

$t_{2n-2, \frac{\alpha}{2}}$ $\equiv$ t-distribution

$$CI(95\%) = [-1.2\%; 2.0\%]$$

$$CI(99\%) = [-1.9\%; 2.7\%]$$

Accepted



$v \backslash \alpha$	0,1	0,05	0,025	0,01	0,005	0,001	0,0005
1	3,08	6,31	12,71	31,82	63,66	318,29	636,58
2	1,89	2,92	4,30	6,96	9,92	22,33	31,60
3	1,64	2,35	3,18	4,54	5,84	10,21	12,92
4	1,53	2,13	2,78	3,75	4,60	7,17	8,61
5	1,48	2,02	2,57	3,36	4,03	5,89	6,87
6	1,44	1,94	2,45	3,14	3,71	5,21	5,96
7	1,41	1,89	2,36	3,00	3,50	4,79	5,41
8	1,40	1,86	2,31	2,90	3,36	4,50	5,04
9	1,38	1,83	2,26	2,82	3,25	4,30	4,78
10	1,37	1,81	2,23	2,76	3,17	4,14	4,59
11	1,36	1,80	2,20	2,72	3,11	4,02	4,44
12	1,36	1,78	2,18	2,68	3,05	3,93	4,32
13	1,35	1,77	2,16	2,65	3,01	3,85	4,22
14	1,35	1,76	2,14	2,62	2,98	3,79	4,14
15	1,34	1,75	2,13	2,60	2,95	3,73	4,07
16	1,34	1,75	2,12	2,58	2,92	3,69	4,01
17	1,33	1,74	2,11	2,57	2,90	3,65	3,97
18	1,33	1,73	2,10	2,55	2,88	3,61	3,92
19	1,33	1,73	2,09	2,54	2,86	3,58	3,88

t-student distribution
[Chwif and Medina, 2006]

$$\bar{x}_s - \bar{x}_r \pm t_{2n-2, \frac{\alpha}{2}} \sqrt{\frac{S_s^2 + S_r^2}{n}}$$

$\bar{x}_s$ $\equiv$ mean of simulation result

$\bar{x}_r$ $\equiv$ mean of real result

$\bar{S}_s$ $\equiv$ std. dev of simulation

$\bar{S}_r$ $\equiv$ std. dev of real

n $\equiv$ data (same for real and sim)

$t_{2n-2, \frac{\alpha}{2}}$ $\equiv$ t-distribution

$$CI(95\%) = [-1.2\%; 2.0\%]$$

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Accepted

Validation without real data

Causal Influence Matrix (CIM)

- +1 : Positive correlation
- 1 : Negative correlation
- 0 : Neutral correlation (*input do not affect output*)

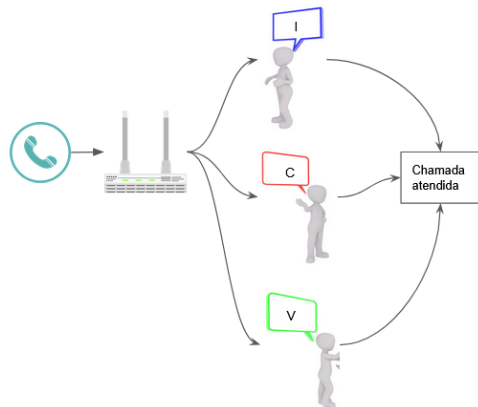
Correlations can be established based on the analyst's experience with the system (maximum of 9 inputs and 9 outputs).

Inputs

- ▶ **I**: Information service
- ▶ **V**: Sales service

Outputs

- ▶ S_1 : Lost calls
- ▶ S_2 : % calls answered within 2 min
- ▶ S_3 : Mean wait time



Example in [Chwif and Medina, 2006]

Validation without real data

CIM (2 input, 3 outputs)

Inputs

- ▶ **I**: Information service
- ▶ **V**: Sales service

	S_1	S_2	S_3
I	-1	+1	-1
V	-1	+1	-1

I : Number of Information services

V: Number of Sales services

Scenarios for input variables: 4 or 8 service units: **(4, 4)**, **(4, 8)**, **(8, 4)**, **(8, 8)**

	(4, 4)	(4, 8)	(8, 4)	(8, 8)
S_1	644 chamadas	397 chamadas	406 chamadas	76 chamadas
S_2	63%	77%	80%	89%
S_3	1,44 minutos	0,96 minutos	0,86 minutos	0,52 minutos

Outputs

- ▶ S_1 : Lost calls
- ▶ S_2 : % calls answered within 2 min
- ▶ S_3 : Mean wait time

Example in [Chwif and Medina, 2006]



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Sizing simulation runs [Banks, 2005, Law et al., 2007, Pidd, 1998]

To set the **number of replications** to achieve a **precision** h^* for a performance measure, follow these steps:

1. Set the desired significance level (usually $\alpha = 0.05$): $P(\bar{x} - h, \bar{x} + h) = 1 - \alpha$
2. Perform a run with n replications ($n \geq 5$).
3. Calculate the precision for the variable (h): $h = t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}}$
4. If $h \leq h^*$, then stop (precision **achieved**).
Else: increase the number of replications to n^* : $n^* = \lceil n(\frac{h}{h^*})^2 \rceil$
5. Set $n = n^*$ and return to **step 2**.

Sizing simulation runs [Banks, 2005, Law et al., 2007, Pidd, 1998]

Example: When simulating 10 replications of a hospital surgical ward's operations, we observed an average of 1.54 patients and a standard deviation of 2.03.

- ▶ **Performance measure:** Average number of patients in line: μ

- ▶ $\bar{x} = 1.54$ Confidence: 95% :
 $P(\bar{x} - h, \bar{x} + h) = 1 - \alpha$. Then
 $\alpha = 0.05$, and $\frac{\alpha}{2} = 0.025$.

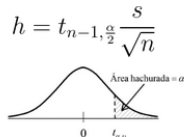
- ▶ Degrees of freedom: $n - 1 = 10 - 1 = 9$.

- ▶ $t_{n-1, \frac{\alpha}{2}} = 2.26$, $\sqrt{10} = 3.16$, $s = 2.03$

- ▶ $h = t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} = 2.26 * \frac{2.03}{3.16} = 1.45$

- ▶ $P(1.54 - 1.45 \leq \mu \leq 1.54 + 1.45) = 1 - 0.05$

- ▶ $P(0.09 \leq \mu \leq 2.99) = 95\%$



$v \backslash \alpha$	0,1	0,05	0,025	0,01	0,005	0,001	0,0005
1	3,08	6,31	12,71	31,82	63,66	318,29	636,58
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Running Terminal Systems [Banks, 2005, Law et al., 2007, Pidd, 1998]

The **duration** of the replications must coincide with the **terminal system** operating hours. The number of replications depends on the desired **precision** and **confidence level**: $P(\bar{x} - h, \bar{x} + h) = 1 - \alpha$.

1. Set the **performance measure** to be analyzed.
2. Set a confidence level (e.g., 95%) and a precision level (e.g., 3% of the mean).
3. Define the precision h^* for the performance measure.
4. Define the duration each replication: the system's operating hours.
5. Perform a pilot run with n replications (e.g., $n = 5$).
6. Calculate the number of replications, n^* to obtain h^* for the performance measure.
7. Run the model with n^* replications to obtain the performance measure results.

Running Non-Terminal Systems [Banks, 2005, Law et al., 2007, Pidd, 1998]

A **warm-up time** is required. The number of replications depends on the desired **precision** and **confidence level**: $P(\bar{x} - h, \bar{x} + h) = 1 - \alpha$.

1. Set the **performance measure** to be analyzed.
2. Set a confidence level (e.g., 95%) and a precision level (e.g., 3% of the mean).
3. Define the precision h^* for the performance measure.
4. Identify a **warm-up** time.
5. Define the duration each replication: the longer, the lower is the std deviation.
6. Perform a pilot run with n replications (e.g., $n = 5$).
7. Calculate the number of replications, n^* to obtain h^* for the perf. measure.
8. Run the model with n^* replications to obtain the performance measure results.

Comparing two scenarios [Banks, 2005, Law et al., 2007, Pidd, 1998]

With **the same (n) number of replications**. Let x_{ij} be value of performance measure from replication i of scenario j

1. Compute differences: $d_i = x_{i1} - x_{i2}$ (scenario 1 - scenario 2)
2. Compute mean and std dev of differences: $\bar{d}_i = \frac{\sum_{i=1}^n d_i}{n}$, and $s = \sqrt{\frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n-1}}$.
3. Compute CI for the mean of differences: $CI = \bar{d} \pm h$, where $h = t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}}$.

Let **CI** = [**a**,**b**], and remember $d_i = x_{i1} - x_{i2}$ (scenario 1 - scenario 2)

- ▶ If CI has a zero, i.e., $a < 0$ and $b > 0$: we **can not say** mean of scenario 1 higher or lower than scenario 2.
- ▶ If CI has $a < 0$ and $b < 0$: mean of **scenario 2 is higher** than of scenario 1.
- ▶ If CI has $a > 0$ and $b > 0$: mean of **scenario 1 is higher** than of scenario 2.

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